

MENG DING, China

WMO: 569450

Lat: 23.567N

Lon: 99.083E

Elev: 512

StdP: 95.32

Time Zone: 8.00 (E08)

Period: 82-89

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.0	(c) 9.0	(d) 5.9	(e) 6.1	(f) 19.4	(g) 6.9	(h) 6.6	(i) 17.0	(j) 5.3	(k) 25.2	(l) 4.5	(m) 24.5	(n) 0.1	(o) N/A	(p) 0.262

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.2	(c) 35.5	(d) 22.3	(e) 34.0	(f) 22.3	(g) 32.7	(h) 22.8	(i) 26.2	(j) 31.1	(k) 25.8	(l) 30.4	(m) 25.4	(n) 29.8	(o) 3.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.2	28.9	24.5	20.8	28.2	24.2	20.3	27.7	85.3	31.4	83.4	30.7	81.7	30.1	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.0	(b) 4.6	(c) 3.5	(e) 6.0	(f) 38.3	(g) 1.1	(h) 0.8	(i) 5.2	(j) 38.9	(k) 4.5	(l) 39.4	(m) 3.9	(n) 39.8	(o) 3.1	(p) 40.4		
(a) 6.0	(b) 4.6	(c) 3.5	(e) 5.8	(f) 28.4	(g) 1.1	(h) 1.6	(i) 5.0	(j) 29.5	(k) 4.4	(l) 30.5	(m) 3.8	(n) 31.4	(o) 3.0	(p) 32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	17.1	19.0	21.9	24.2	26.8	26.9	26.2	26.7	26.1	24.4	20.6	17.4		
	DBStd	3.99	1.46	1.73	2.14	1.87	2.14	1.64	1.61	1.22	1.30	1.72	1.67	1.66		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	99	43	12	3	0	0	0	0	0	0	0	2	39		
	CDD10.0	4793	220	252	369	426	521	508	502	516	484	447	319	229		
	CDD18.3	1852	4	31	114	176	263	258	243	258	234	188	71	10		
	CDH23.3	14897	84	378	1239	1553	2557	2267	1827	2002	1623	1048	269	48		
	CDH26.7	5273	3	87	509	697	1251	773	506	597	515	301	32	1		
(14) Wind	WSAvg	0.9	0.6	0.9	1.1	1.2	1.3	1.1	1.0	1.0	0.9	0.6	0.5	0.4		
(15) Precipitation	PrecAvg	1519	14	18	14	44	127	276	339	314	169	124	67	16		
	PrecMax	1820	71	77	55	151	338	541	497	482	377	318	210	70		
	PrecMin	1250	0	0	0	1	30	111	179	126	69	44	0	1		
	PrecStd	150	16	19	16	29	79	91	85	88	70	74	56	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	30.8	34.0	35.9	37.6	36.5	33.5	33.1	33.4	32.0	29.1	26.2		
		MCWB	17.1	17.4	18.9	19.8	22.1	24.6	26.3	25.8	25.7	24.6	21.5	18.6		
	2%	DB	25.6	29.2	32.5	34.2	36.2	33.3	32.0	32.0	32.1	30.8	27.6	24.7		
		MCWB	16.4	17.0	18.6	20.0	22.6	25.1	25.9	25.7	25.4	24.0	20.2	17.7		
	5%	DB	24.1	27.8	31.3	32.9	34.8	31.8	30.6	31.0	30.9	29.7	26.3	23.4		
		MCWB	16.0	16.6	18.3	19.9	22.6	24.9	25.3	25.5	25.0	23.6	19.7	17.0		
	10%	DB	22.6	26.0	29.8	30.9	32.9	30.5	29.5	29.9	29.6	28.5	24.9	21.9		
		MCWB	15.5	16.2	17.8	19.9	22.7	24.8	25.0	25.2	24.7	23.0	19.3	16.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	18.5	20.6	22.7	25.3	26.5	26.7	26.7	26.6	25.5	22.5	19.4		
		MCDB	24.8	28.1	30.4	29.7	31.2	31.8	32.3	31.5	31.4	30.4	26.2	24.7		
	2%	WB	17.2	17.7	19.8	22.0	24.6	25.8	26.1	26.1	25.8	24.7	21.5	18.3		
		MCDB	23.6	26.8	29.8	28.5	30.5	30.9	30.9	30.7	30.6	29.3	24.5	23.4		
	5%	WB	16.5	17.1	19.1	21.4	24.0	25.4	25.6	25.7	25.3	24.1	20.9	17.5		
		MCDB	22.7	25.7	28.9	28.2	30.0	30.0	29.8	30.0	29.7	28.3	24.3	22.0		
	10%	WB	15.8	16.5	18.4	20.9	23.5	25.0	25.1	25.3	24.8	23.6	20.2	16.7		
		MCDB	21.7	24.5	27.8	27.8	29.5	29.2	28.8	29.2	28.8	27.2	23.6	20.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	16.1	16.9	13.5	11.6	7.2	6.0	7.0	8.1	9.3	10.6	12.3		
		MCDBR	16.1	18.8	19.1	17.5	15.0	9.9	8.5	8.8	10.3	10.9	12.4	13.2		
	5% WB	MCWBR	7.1	7.0	6.5	5.6	3.8	3.0	3.0	3.4	4.0	4.4	5.2	6.0		
		MCWBR	14.7	16.8	16.9	12.9	10.5	7.6	7.6	8.0	8.9	9.3	9.7	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.375	0.438	0.710	0.704	0.538	0.470	0.459	0.497	0.489	0.465	0.404	0.378		
	taud		2.327	2.144	1.513	1.588	2.035	2.285	2.311	2.179	2.177	2.217	2.326	2.354		
	Ebn at Noon		881	843	643	662	779	827	836	807	807	808	844	863		
	Edh at Noon		115	147	290	274	174	135	131	150	147	135	114	108		
(44) All-Sky Solar Radiation	RadAvg		4.36	5.23	5.42	5.46	4.92	4.20	3.85	4.25	4.29	4.08	4.21	3.88		
	RadStd		0.31	0.38	0.39	0.41	0.51	0.39	0.27	0.33	0.30	0.43	0.31	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.25	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	+92	+55

Nomenclature: See separate page